PenicillinX-Project

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Hutten et al.

Figure 1 is a reconstruction of the matrix detailed in Hutten et al[1] Below them are the various molecular matching algorithms and their respective matrices, the performance of each algorithm, as well as their combined performance matching against Hutten et al is displayed in a table at the end of the document. Matrices generated using R library complexheatmap[2,3]. *This matrix was reconstructed by hand, and though it was checked for consistency against the original, it can't be guaranteed that it is free from transcription error.*

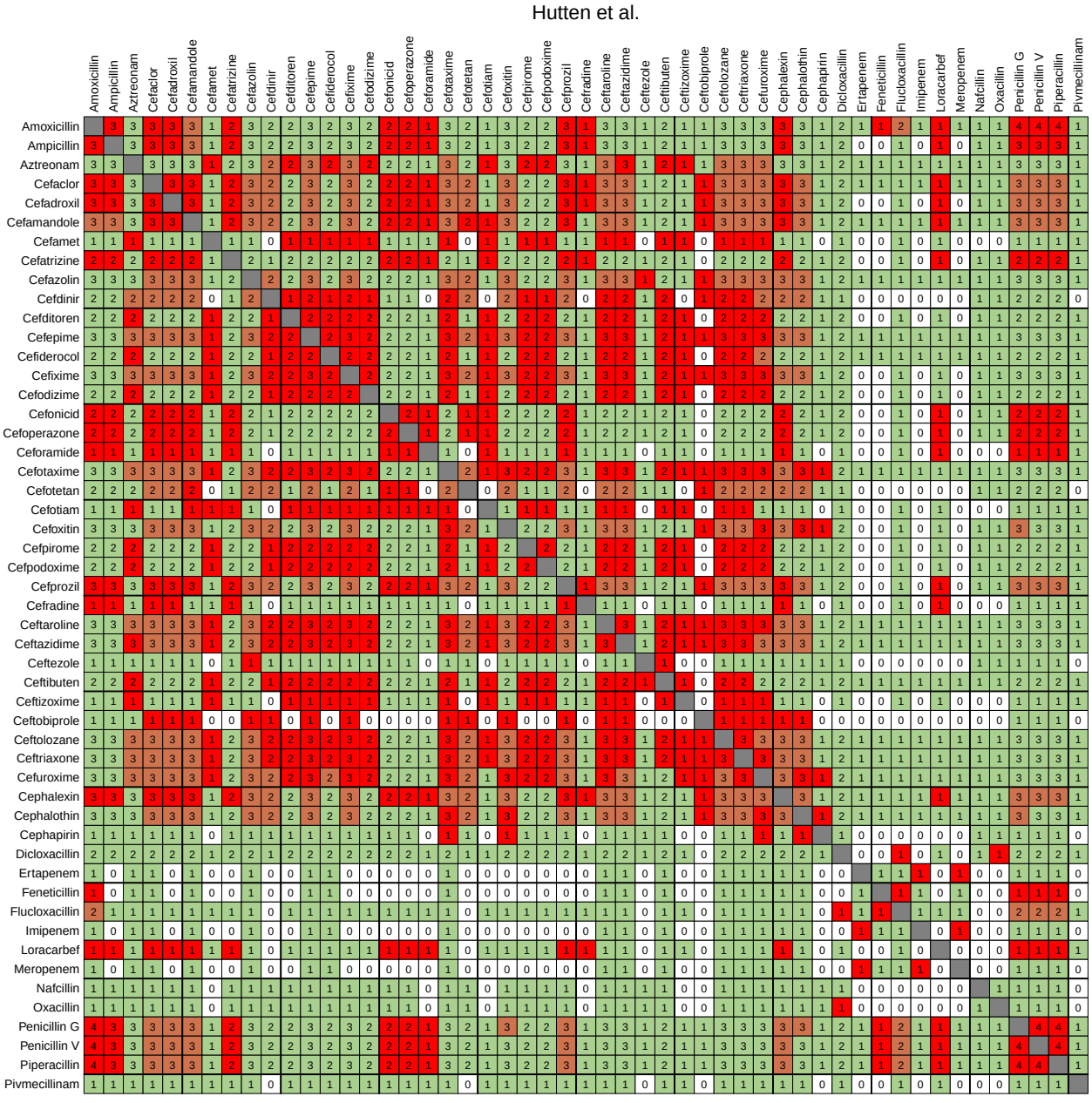

Figure 1: Cross reactivity matrix manually re-created from Hutten et al.

Direct Matching (PenicillinX Original)

The existing direct matching algorithm from PenicillinX web app.[4] Compares molecular structures directly using graph based comparisons of side chain SMILES. Sensitivity is limited, and often does not account for R1 alterations in side chain moieties that do not change allergenic potential.

Figure 2: Cross reactivity matrix using PenicillinX direct matching algorithm

Permissive Direct Matching (PDM) using SMARTS substitutions

A Matching algorithm similar to original PenicillinX,[4] but instead using SMARTS pattern matching.[5] It implements bioisosteric substitution for features with similar expected behaviour specifically treating halogens as a united class (i.e. flucloxacillin-dicloxacillin) as well as methylene and ether linkages. (i.e. Penicillin G-VK). Continued work on this method shows promise in increasing the sensitivity of the algorithm while minimising loss to specificity.

Figure 3: Cross reactivity matrix using an improved, more pemissive matching algorithm

Dice Similarity with Morgan Fingerprinting

This method utilises Morgan circular fingerprints,[6] which iteratively encodes the local atomic environment into a bit vector. These fingerprints are then compared between two side chains using the Dice Coefficient. The Dice coefficient assigns values to molecular structures, and weighs shared features more heaviliy than the total feature set, providing a measure of structural overlap.

$$S_{AB} = \frac{2C}{A+B} \tag{1}$$

where S_{AB} is similarity coefficient, C is the common features between both molecules, and A & B are the features in each molecule.

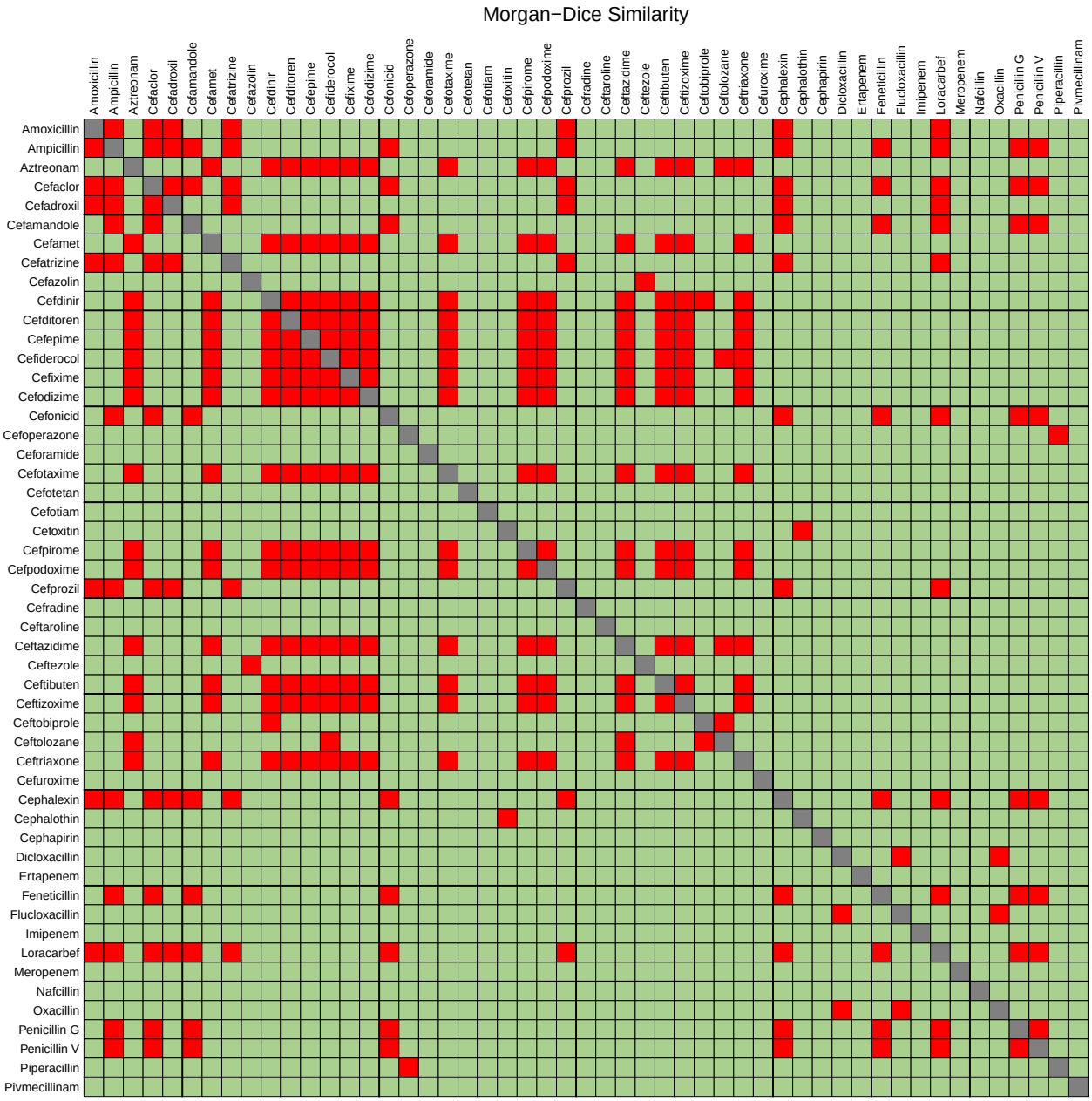

Figure 4: Cross reactivity matrix using Morgan fingerprinting with Dice Similarity

Tanimoto Similarity with Morgan Fingerprinting

A variation of the previous algorithm that instead utilises the Tanimoto Coefficient. Similar in concept to the Dice Coefficient, but with a different weight applied to matching substructures. Previous modelling in this project indicated that the Dice Coefficient strictly dominates the Tanimoto Coefficient, in detecting cross reactivity. Thus analyses using Tanimoto have been excluded from the final analysis.

$$S_{AB} = \frac{C}{A + B - C} \tag{2}$$

Figure 5: Cross reactivity matrix using Morgan fingerprinting with Tanimoto Similarity

Pharmacocore with Dice Similarity

Pharmacocore fingerprinting maps the chemical characteristics of atoms and functional groups, such as hydrogen bond acceptors/donors, aliphatics/aromtatics. The Dice coefficient is then applied to attempt to model how an IgE molecule may 'perceive' a side chain's binding potential regardless of non-functional differences in the molecular structure.[7]

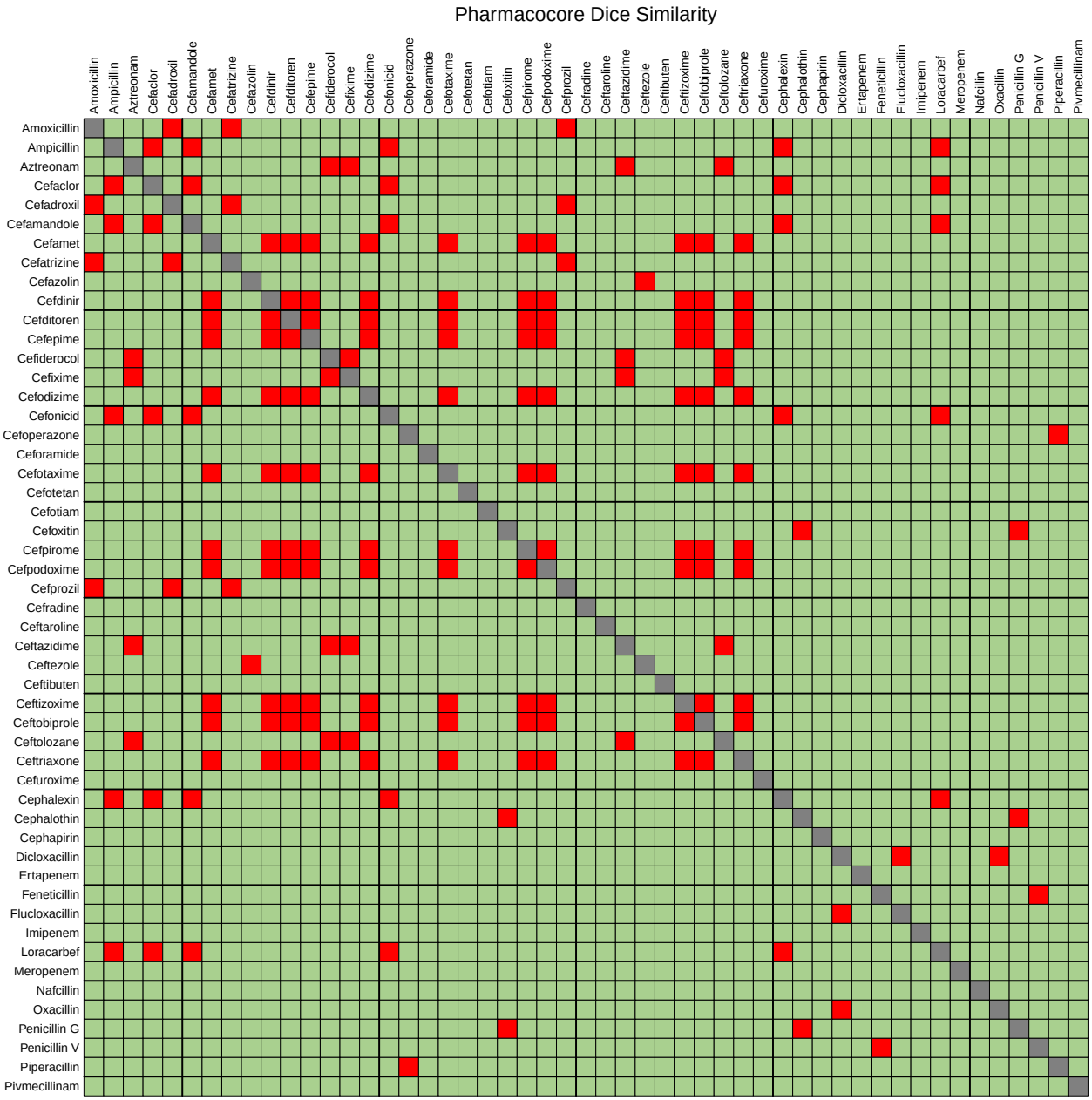

Figure 6: Cross reactivity matrix using Morgan fingerprinting with Pharmacocores

MACCS with Dice Similarity

This method utilises Molecular Access System(MACCS) keys, a specific type of pre-defined, publicly available structural fragment definition. There are 166 public fragment definitions in the RDKit implementation. A bit string is generated with reference to the presence or absence of certain features, for example, < 3 oxygens, S-S bond, or ring of size 4, etc.[9] This method attempts to capture broad structural classes and motifs rather than atom-by-atom circular fingerprints.

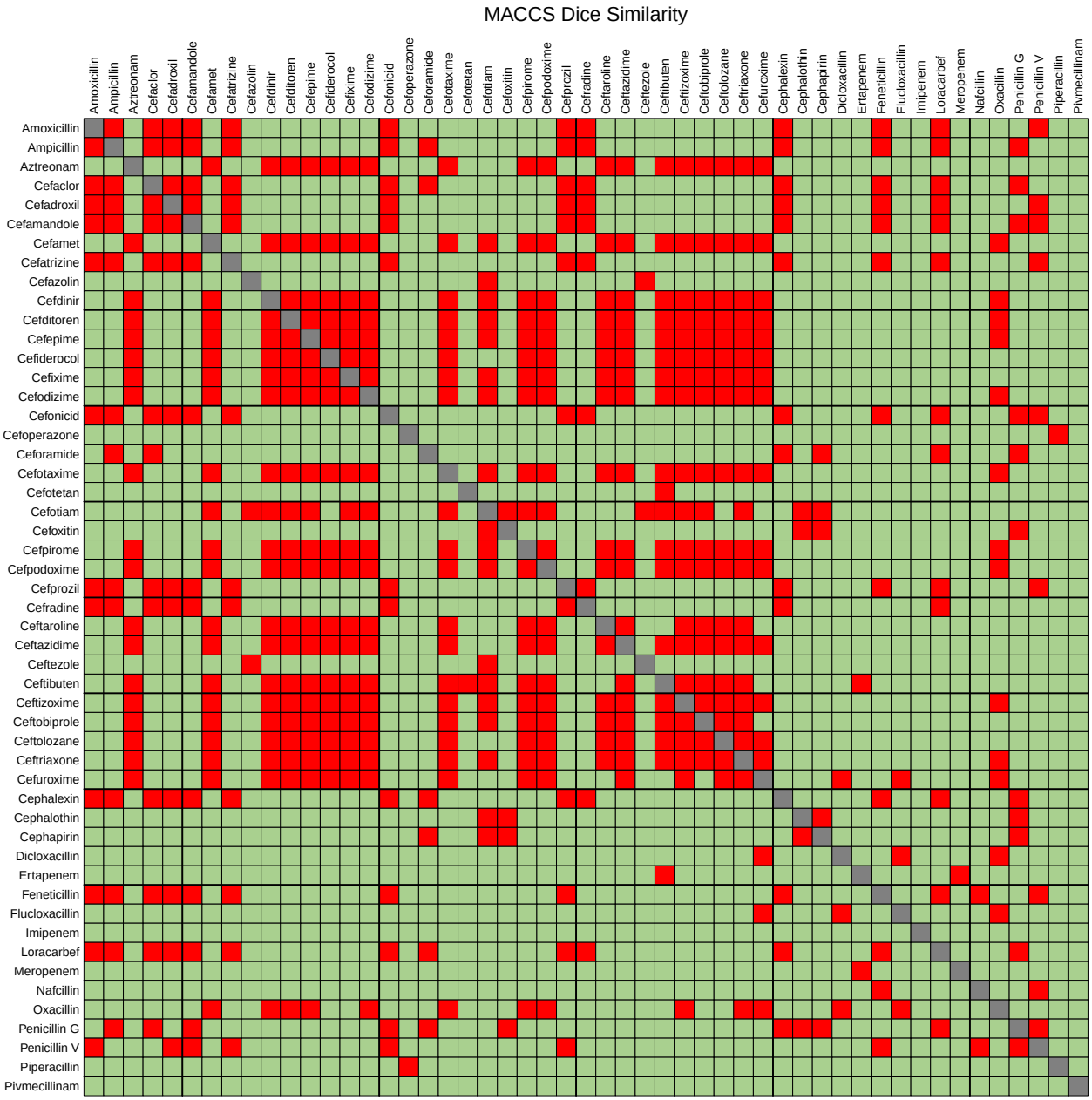

Figure 7: Cross reactivity matrix using MACCS keys with Dice Similarity

Maximum Common Substructure (MCS) Overlap

The MCS algorithm identifies the largest substructure shared by two side chains. It performs a graph-matching search to find the maximum number of identical atoms and bonds common to both molecules. A similarity score is calculated as the ratio of the shared atoms to the total atoms in the smaller of the two molecules.[10] Calculated as:

$$S_{MCS} = \frac{|MCS(A,B)|}{\min(|A|,|B|)} \tag{3}$$

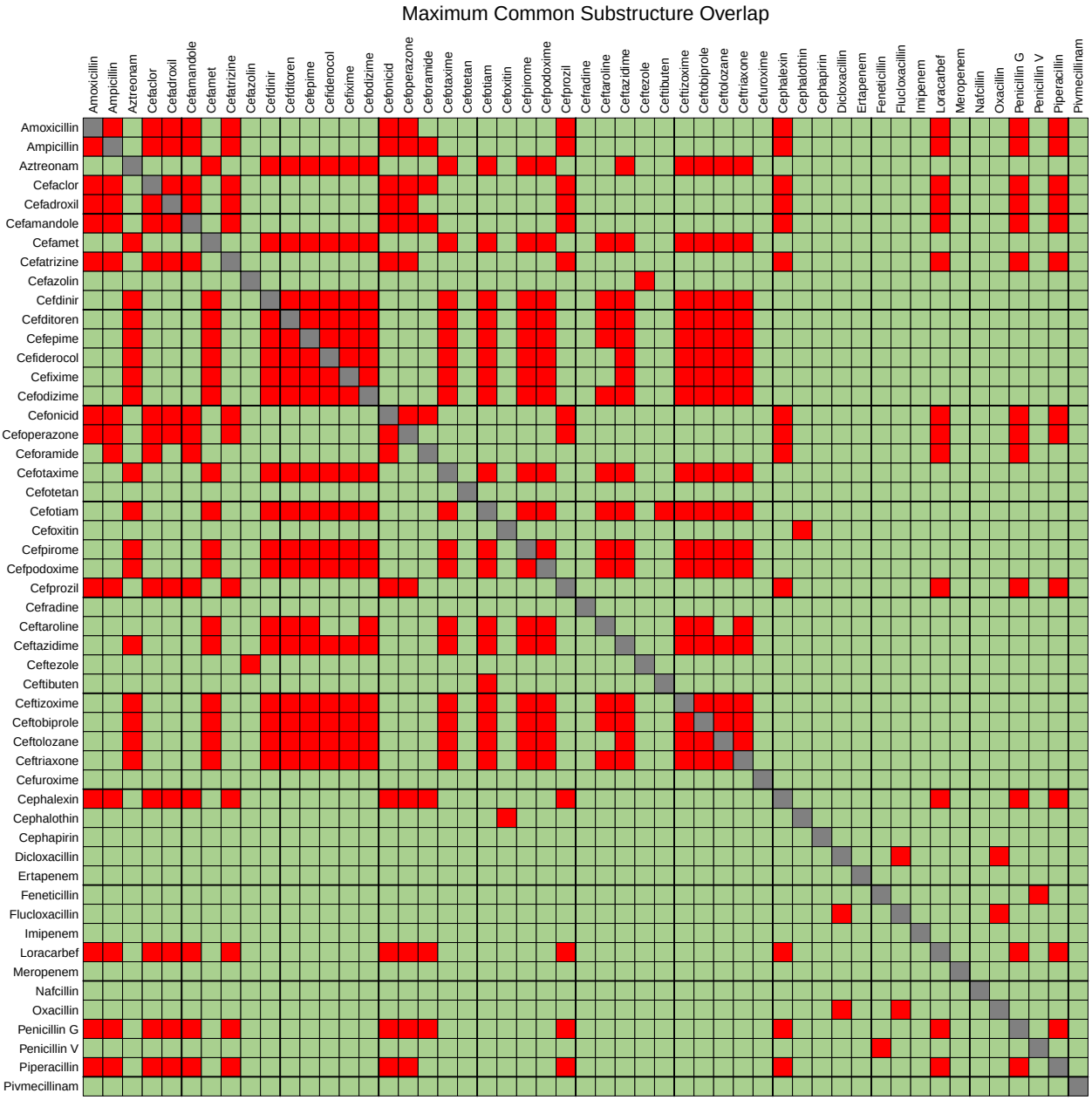

Figure 8: Cross reactivity matrix using Maximum Common Substructure

All algorithms combined

A matrix generated with the combined hits of all previous algorithms, currently excluding Morgan Tanimoto due to poor performance (it does not find matches not already covered by other algorithms). MCS overlap has been re-added after further tweaking, having been removed in earlier iterations due to poor sensitivity.

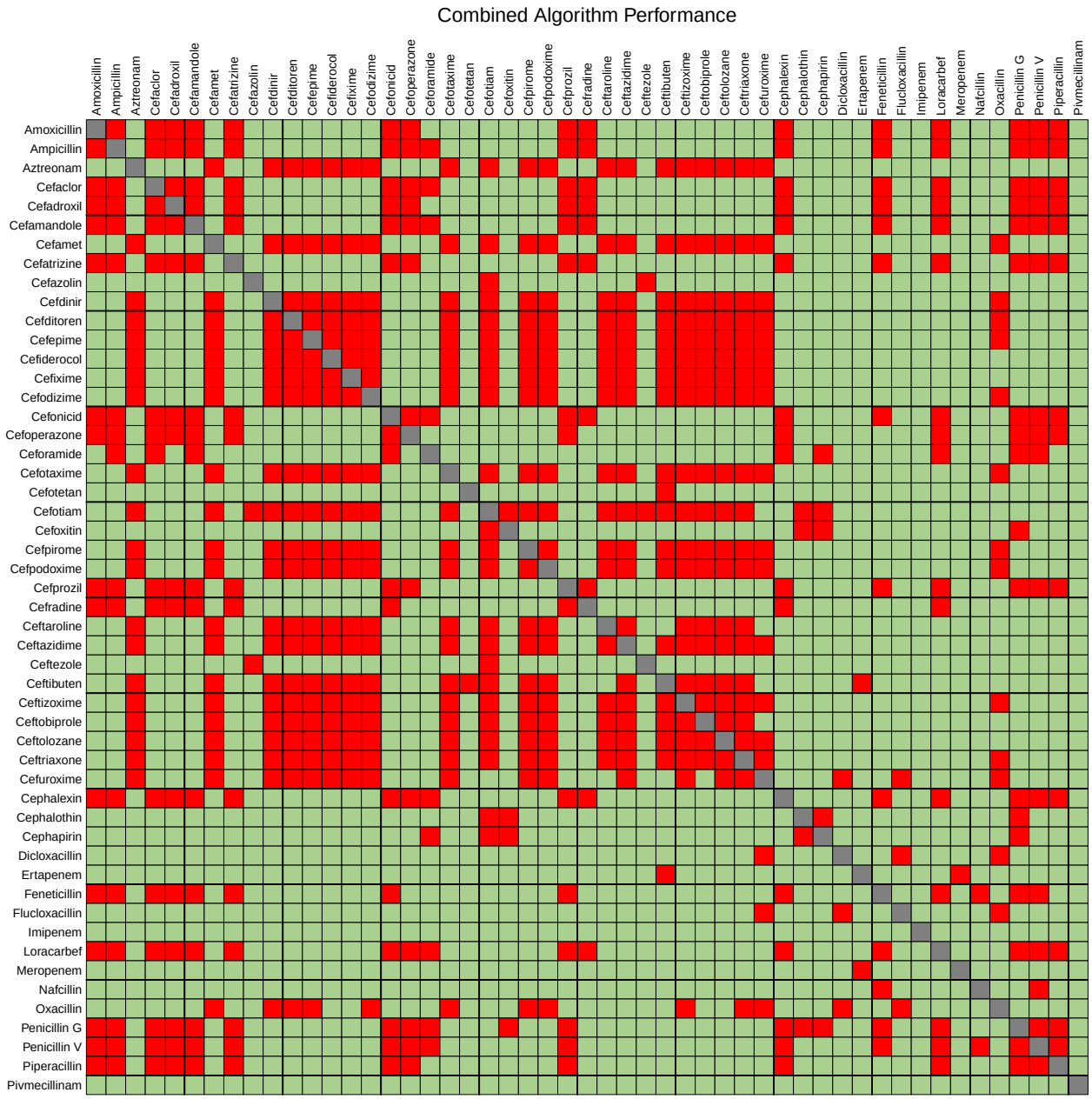

Figure 9: Combined performance of all algorithms

Hutten et al. without discrepants

A re-creation of the Hutten et al[1] Matrix with missing values and discrepant matches changed from red and white to green respectively. Currently present purely to aid in visual comparisons.

Figure 10: Hutten with discrepants and missing values set to green for better visual comparison

Performance Table

A table detailing the performance of each algorithm with the final column demonstrating the performance of all algorithms combined. Youden's J is used as an overall performance metric to assist in gauging the cost/benefit of each change to an algorithm. Youden's J is calculated as:

$$J = \text{Sensitivity} + \text{Specificity} - 1 \tag{4}$$

I expect there should be a way that I can automate the process of tweaking the thresholds for the similarity coefficients to achieve an optimum Youden's J. Thus far these tweaks have been done manually, then revised by observing the result.

Table 1: Compared performance of each algorithm Morgan Tanimoto not included in combined result

Metric	Direct Match	Morgan Dice	PDM	Ph-Dice	MCS	MACCS Dice	Combo
Threshold	N/A	0.55	N/A	0.67	0.89	0.68	N/A
Possible Hits	297.0	297	297.0	297	297	297	297.0
Total Hits	168.0	174	183.0	94	240	299	340.0
True +ve	152.0	152	161.0	81	209	228	260.0
False +ve	0.0	3	1.0	0	1	24	24.0
False -ve	145.0	145	136.0	216	88	69	37.0
no Data	4.0	6	4.0	8	12	25	25.0
Sensitivity	0.512	0.512	0.542	0.273	0.704	0.768	0.875
Specificity	1.0	0.995	0.998	1	0.998	0.964	0.964
Youden's J	0.512	0.507	0.541	0.273	0.702	0.732	0.839

Raw Results

A generated list of matches deemed to be false positives or negatives when compared to the Hutten et al. chart for auditing purposes. These can be plugged into PenicillinX[4] for visual inspection of the structures in order to generate hypotheses on why a particular algorithm may have incorrectly hit or missed potential interactions.

25 Predicted cross-reactions with no clinical data in Hutten et al.

- Ampicillin and Feneticillin [**Morgan, MACCS**]
- Cefadroxil and Feneticillin [**MACCS**]
- Cefamet and Cefdinir [**DM, PDM, Morgan, Pharmaco, MCS, MACCS**]
- Cefamet and Ceftobiprole [**Pharmaco, MCS, MACCS**]
- Cefamet and Oxacillin [**MACCS**]
- Cefatrizine and Feneticillin [**MACCS**]

- Cefdinir and Cefotiam [**DM, PDM, MCS, MACCS**]
- Cefdinir and Ceftizoxime [**DM, PDM, Morgan, Pharmaco, MCS, MACCS**]
- Cefditoren and Ceftobiprole [**Pharmaco, MCS, MACCS**]
- Cefiderocol and Ceftobiprole [**MCS, MACCS**]
- Cefodizime and Ceftobiprole [**Pharmaco, MCS, MACCS**]
- Cefonicid and Feneticillin [**Morgan, MACCS**]
- Ceforamide and Cephapirin [**MACCS**]
- Cefotiam and Ceftezole [**MACCS**]
- Cefotiam and Ceftobiprole [**MCS, MACCS**]
- Cefotiam and Cephapirin [**MACCS**]
- Cefpirome and Ceftobiprole [**Pharmaco, MCS, MACCS**]
- Cefpodoxime and Ceftobiprole [**Pharmaco, MCS, MACCS**]
- Cefprozil and Feneticillin [**MACCS**]
- Ceftibuten and Ceftobiprole [**MACCS**]
- Ceftizoxime and Ceftobiprole [**Pharmaco, MCS, MACCS**]
- Ceftizoxime and Oxacillin [**MACCS**]
- Feneticillin and Loracarbef [**Morgan, MACCS**]
- Feneticillin and Nafcillin [**MACCS**]
- Flucloxacillin and Oxacillin [**DM, PDM, Morgan, MCS, MACCS**]

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24 Potential false positives as compared to Hutten et al.

- Aztreonam and Ceftobiprole [**MCS, MACCS**]
- Cefaclor and Feneticillin [**Morgan, MACCS**]
- Cefamandole and Cefradine [**MACCS**]
- Cefamandole and Feneticillin [**Morgan, MACCS**]
- Cefazolin and Cefotiam [**MACCS**]
- Cefdinir and Oxacillin [**MACCS**]
- Cefditoren and Oxacillin [**MACCS**]
- Cefepime and Oxacillin [**MACCS**]
- Cefodizime and Oxacillin [**MACCS**]
- Cefonicid and Cefradine [**MACCS**]
- Cefotaxime and Oxacillin [**MACCS**]

- Cefotetan and Ceftibuten [**MACCS**]
- Cefotiam and Cefoxitin [**MACCS**]
- Cefotiam and Cephalothin [**MACCS**]
- Cefpirome and Oxacillin [**MACCS**]
- Cefpodoxime and Oxacillin [**MACCS**]
- Ceftibuten and Ertapenem [**MACCS**]
- Ceftriaxone and Oxacillin [**MACCS**]
- Cefuroxime and Dicloxacillin [**MACCS**]
- Cefuroxime and Flucloxacillin [**MACCS**]
- Cefuroxime and Oxacillin [**MACCS**]
- Cephalexin and Feneticillin [**Morgan, MACCS**]
- Cephapirin and Penicillin G [**MACCS**]
- Nafcillin and Penicillin V [**PDM, MACCS**]

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31 Predicted cross-reactions where there is discrepancy according to Hutten et al.

- Amoxicillin and Cefamandole [**MCS, MACCS**]
- Ampicillin and Cefamandole [**Morgan, Pharmaco, MCS, MACCS**]
- Aztreonam and Cefdinir [**DM, PDM, Morgan, MCS, MACCS**]
- Aztreonam and Cefepime [**DM, PDM, Morgan, MCS, MACCS**]
- Aztreonam and Cefixime [**DM, PDM, Morgan, Pharmaco, MCS, MACCS**]
- Aztreonam and Cefotaxime [**DM, PDM, Morgan, MCS, MACCS**]
- Aztreonam and Ceftaroline [**MACCS**]
- Aztreonam and Ceftolozane [**Morgan, Pharmaco, MCS, MACCS**]
- Aztreonam and Ceftriaxone [**DM, PDM, Morgan, MCS, MACCS**]
- Aztreonam and Cefuroxime [**MACCS**]
- Cefaclor and Penicillin G [**DM, PDM, Morgan, MCS, MACCS**]
- Cefaclor and Penicillin V [**PDM, Morgan**]
- Cefaclor and Piperacillin [**DM, PDM, MCS**]
- Cefadroxil and Penicillin G [**DM, PDM, MCS**]
- Cefadroxil and Penicillin V [**PDM, MACCS**]
- Cefadroxil and Piperacillin [**MCS**]
- Cefamandole and Penicillin G [**DM, PDM, Morgan, MCS, MACCS**]

- Cefamandole and Penicillin V [**PDM, Morgan, MACCS**]
- Cefamandole and Piperacillin [**MCS**]
- Cefdinir and Cefuroxime [**MACCS**]
- Cefiderocol and Cefuroxime [**MACCS**]
- Cefoxitin and Penicillin G [**Pharmaco, MACCS**]
- Cefprozil and Penicillin G [**DM, PDM, MCS**]
- Cefprozil and Penicillin V [**PDM, MACCS**]
- Cefprozil and Piperacillin [**MCS**]
- Ceftazidime and Cefuroxime [**MACCS**]
- Ceftolozane and Cefuroxime [**MACCS**]
- Cephalexin and Penicillin G [**DM, PDM, Morgan, MCS, MACCS**]
- Cephalexin and Penicillin V [**PDM, Morgan**]
- Cephalexin and Piperacillin [**DM, PDM, MCS**]
- Cephalothin and Penicillin G [**Pharmaco, MACCS**]

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105 Negative predictions where there is discrepancy according to Hutten et al.

- Amoxicillin and Flucloxacillin [None]
- Cefaclor and Cefazolin [None]
- Cefaclor and Cefdinir [None]
- Cefaclor and Cefepime [None]
- Cefaclor and Cefixime [None]
- Cefaclor and Cefotaxime [None]
- Cefaclor and Cefotetan [None]
- Cefaclor and Cefoxitin [None]
- Cefaclor and Ceftaroline [None]
- Cefaclor and Ceftazidime [None]
- Cefaclor and Ceftolozane [None]
- Cefaclor and Ceftriaxone [None]
- Cefaclor and Cefuroxime [None]
- Cefaclor and Cephalothin [None]
- Cefadroxil and Cefazolin [None]
- Cefadroxil and Cefdinir [None]

- Cefadroxil and Cefepime [None]
- Cefadroxil and Cefixime [None]
- Cefadroxil and Cefotaxime [None]
- Cefadroxil and Cefotetan [None]
- Cefadroxil and Cefoxitin [None]
- Cefadroxil and Ceftaroline [None]
- Cefadroxil and Ceftazidime [None]
- Cefadroxil and Ceftolozane [None]
- Cefadroxil and Ceftriaxone [None]
- Cefadroxil and Cefuroxime [None]
- Cefadroxil and Cephalothin [None]
- Cefamandole and Cefazolin [None]
- Cefamandole and Cefdinir [None]
- Cefamandole and Cefepime [None]
- Cefamandole and Cefixime [None]
- Cefamandole and Cefotaxime [None]
- Cefamandole and Cefoxitin [None]
- Cefamandole and Ceftaroline [None]
- Cefamandole and Ceftazidime [None]
- Cefamandole and Ceftolozane [None]
- Cefamandole and Ceftriaxone [None]
- Cefamandole and Cefuroxime [None]
- Cefamandole and Cephalothin [None]
- Cefazolin and Cefdinir [None]
- Cefazolin and Cefepime [None]
- Cefazolin and Cefixime [None]
- Cefazolin and Cefotaxime [None]
- Cefazolin and Cefotetan [None]
- Cefazolin and Cefoxitin [None]
- Cefazolin and Cefprozil [None]
- Cefazolin and Ceftaroline [None]
- Cefazolin and Ceftazidime [None]

- Cefazolin and Ceftolozane [None]
- Cefazolin and Ceftriaxone [None]
- Cefazolin and Cefuroxime [None]
- Cefazolin and Cephalexin [None]
- Cefazolin and Cephalothin [None]
- Cefdinir and Cefotetan [None]
- Cefdinir and Cefoxitin [None]
- Cefdinir and Cefprozil [None]
- Cefdinir and Cephalexin [None]
- Cefdinir and Cephalothin [None]
- Cefepime and Cefotetan [None]
- Cefepime and Cefoxitin [None]
- Cefepime and Cefprozil [None]
- Cefepime and Cephalexin [None]
- Cefepime and Cephalothin [None]
- Cefixime and Cefotetan [None]
- Cefixime and Cefoxitin [None]
- Cefixime and Cefprozil [None]
- Cefixime and Cephalexin [None]
- Cefixime and Cephalothin [None]
- Cefotaxime and Cefotetan [None]
- Cefotaxime and Cefprozil [None]
- Cefotaxime and Cephalexin [None]
- Cefotetan and Cefoxitin [None]
- Cefotetan and Cefprozil [None]
- Cefotetan and Ceftaroline [None]
- Cefotetan and Ceftazidime [None]
- Cefotetan and Ceftolozane [None]
- Cefotetan and Ceftriaxone [None]
- Cefotetan and Cefuroxime [None]
- Cefotetan and Cephalexin [None]
- Cefotetan and Cephalothin [None]

- Cefoxitin and Cefprozil [None]
- Cefoxitin and Ceftaroline [None]
- Cefoxitin and Ceftazidime [None]
- Cefoxitin and Ceftolozane [None]
- Cefoxitin and Ceftriaxone [None]
- Cefoxitin and Cephalexin [None]
- Cefprozil and Ceftaroline [None]
- Cefprozil and Ceftazidime [None]
- Cefprozil and Ceftolozane [None]
- Cefprozil and Ceftriaxone [None]
- Cefprozil and Cefuroxime [None]
- Cefprozil and Cephalothin [None]
- Ceftaroline and Cephalexin [None]
- Ceftaroline and Cephalothin [None]
- Ceftazidime and Cephalexin [None]
- Ceftazidime and Cephalothin [None]
- Ceftolozane and Cephalexin [None]
- Ceftolozane and Cephalothin [None]
- Ceftriaxone and Cephalexin [None]
- Ceftriaxone and Cephalothin [None]
- Cefuroxime and Cephalexin [None]
- Cephalexin and Cephalothin [None]
- Flucloxacillin and Penicillin G [None]
- Flucloxacillin and Penicillin V [None]
- Flucloxacillin and Piperacillin [None]

37 Potential false negatives as compared to Hutten et al.

- Amoxicillin and Ceforamide [None]
- Cefaclor and Ceftobiprole [None]
- Cefadroxil and Ceforamide [None]
- Cefadroxil and Ceftobiprole [None]
- Cefamandole and Cefotetan [None]
- Cefamandole and Cefotiam [None]
- Cefamandole and Ceftobiprole [None]
- Cefatrizine and Ceforamide [None]
- Cefatrizine and Cefotiam [None]
- Cefazolin and Ceftobiprole [None]
- Cefonicid and Cefotetan [None]
- Cefonicid and Cefotiam [None]
- Cefoperazone and Ceforamide [None]
- Cefoperazone and Cefotetan [None]
- Cefoperazone and Cefotiam [None]
- Ceforamide and Cefotiam [None]
- Ceforamide and Cefprozil [None]
- Ceforamide and Piperacillin [None]
- Cefotaxime and Cefoxitin [None]
- Cefotaxime and Cephalothin [None]
- Cefotaxime and Cephapirin [None]
- Cefotetan and Ceftobiprole [None]
- Cefoxitin and Ceftobiprole [None]
- Cefoxitin and Cefuroxime [None]
- Cefprozil and Ceftobiprole [None]
- Ceftaroline and Ceftibuten [None]
- Ceftaroline and Cefuroxime [None]
- Ceftezole and Ceftibuten [None]
- Ceftobiprole and Cefuroxime [None]
- Ceftobiprole and Cephalexin [None]

- Ceftobiprole and Cephalothin [None]
- Cefuroxime and Cephalothin [None]
- Cefuroxime and Cephapirin [None]
- Ertapenem and Imipenem [None]
- Feneticillin and Flucloxacillin [None]
- Feneticillin and Piperacillin [None]
- Imipenem and Meropenem [None]

References

1. Hutten EM, Bulatović Ćalasan M, Trubiano JA, Pleijhuis RG, Terreehorst I. Discrepancies in Beta-Lactam Antibiotics Cross-Reactivity: Implications for Clinical Practice. *Allergy*. 2025;80(11):3108-3114. doi:10.1111/all.16485

2. Gu Z. Complex heatmap visualization. *iMeta*. 2022;1(3):e43. doi:10.1002/imt2.43

3. Gu Z, Gu L, Eils R, Schlesner M, Brors B. *Circlize* implements and enhances circular visualization in R. *Bioinformatics*. 2014;30(19):2811-2812. doi:10.1093/bioinformatics/btu393

4. Liam Jones. PenicillinX. 2025. Accessed January 28, 2026. https://penicillinx.liamjones.science/

5. Daylight Chemical Information Systems, Inc. SMARTS - A Language for Describing Molecular Patterns. In: *Daylight Theory Manual*. 4.9. 11 AD. Accessed January 26, 2026. https://www.daylight.com/dayhtml/doc/theory/theory.smarts.html

6. Morgan HL. The Generation of a Unique Machine Description for Chemical Structures-A Technique Developed at Chemical Abstracts Service. *J Chem Doc*. 1965;5(2):107-113. doi:10.1021/c160017a018

7. Cereto-Massagué A, Ojeda MJ, Valls C, Mulero M, Garcia-Vallvé S, Pujadas G. Molecular fingerprint similarity search in virtual screening. *Methods*. 2015;71:58-63. doi:10.1016/j.ymeth.2014.08.005

8. Greg Landrum. Rdkit/rdkit/Chem/MACCSkeys.py at master · rdkit/rdkit. Published online April 28, 2023. Accessed January 27, 2026. https://github.com/rdkit/rdkit/blob/master/rdkit/Chem/MACCSkeys.py

9. José Aniceto. MACCS fingerprints — Machine Learning Knowledge Base. 2023. Accessed January 27, 2026. https://janiceto.github.io/ml-knowledge-base/02-data-preparation/feature-engineering/maccs.html

10. Andrew Dalke. Rdkit.Chem.fmcs.fmcs module — The RDKit 2025.09.4 documentation. 2012. Accessed January 28, 2026. https://www.rdkit.org/docs/source/rdkit.Chem.fmcs.fmcs.html